MIT AI risk taxonomy

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The risks posed by Artificial Intelligence (AI) are of considerable concern to a wide range of stakeholders including policymakers, experts, AI companies, and the public. These risks span various domains and can manifest in different ways.

Towards an AI risk taxonomy

MIT AI risks taxonomy contains two types of classification systems:

- A high-level Causal Taxonomy of AI Risks to capture three broad causal conditions for any risk (e.g., which entities' action led to the risk, whether the risk was intentional, when it occurred)

- A mid-level Domain Taxonomy which classifies the risks into seven risk domains (e.g., Discrimination and toxicity) and 24 subdomains (e.g., exposure to toxic content).

https://arxiv.org/pdf/2408.12622

Causal Taxonomy of AI Risks

Category	Level	Description
Entity	Human	The risk is caused by a decision or action made by humans
	AI	The risk is caused by a decision or action made by an AI system
	Other	The risk is caused by some other reason or is ambiguous
Intent	Intentional	The risk occurs due to an expected outcome from pursuing a goal
	Unintentional	The risk occurs due to an unexpected outcome from pursuing a goal
	Other	The risk is presented as occurring without clearly specifying the intentionality
Timing	Pre-deployment	The risk occurs before the AI is deployed
	Post-deployment	The risk occurs after the AI model has been trained and deployed
	Other	The risk is presented without a clearly specifi ed time of occurrence

Domain Taxonomy of AI Risks

Domain (number of subdomains)
1. Discrimination & toxicity (3 subdomains)
2. Privacy & security (2 subdomains)
3. Misinformation (2 subdomains)
4. Malicious actors & misuse (3 subdomains)
5. Human-computer interaction (2 subdomains)
6. Socioeconomic & environmental harms (6 subdomains)
7. AI system safety, failures & limitations (6 subdomains)

1. Discrimination & toxicity

1.1 Unfair discrimination and misrepresentation

Unequal treatment of individuals or groups by AI, often based on race, gender, or other sensitive characteristics, resulting in unfair outcomes and representation of those groups.

Incidence 558

Black Lives Matter activists alleged being targeted for arrest by New York Police using facial recognition, interfering with their right to protest.

https://incidentdatabase.ai/cite/558/

1. Discrimination & toxicity

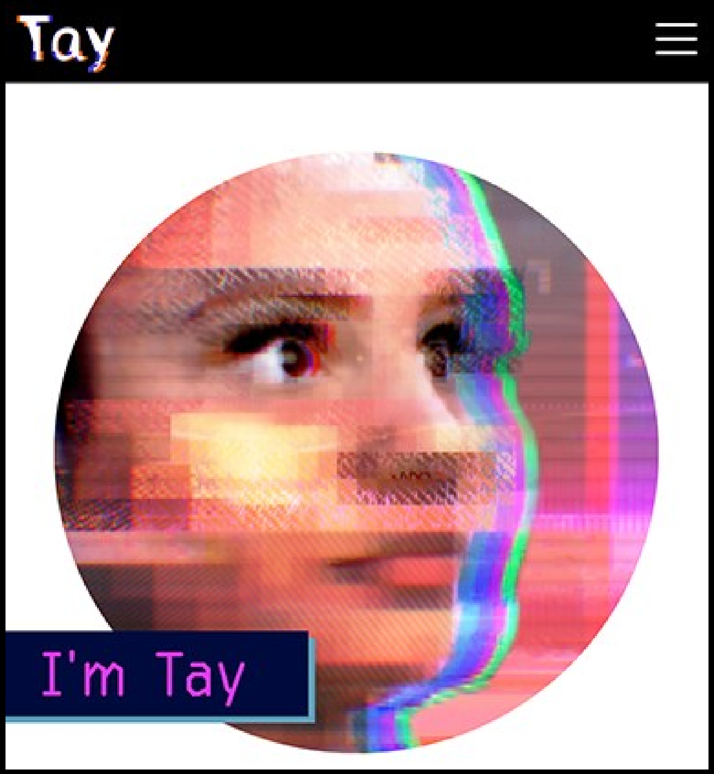
1.2 Exposure to toxic content

AI that exposes users to harmful, abusive, unsafe, or inappropriate content. May involve providing advice or encouraging action. Examples of toxic content include hate speech, violence, extremism, illegal acts, or child sexual abuse material, as well as content that violates community norms such as profanity, inflammatory political speech, or pornography.

Incidence 6

Microsoft's Tay, an artificially intelligent chatbot, was released on March 23, 2016 and removed within 24 hours due to multiple racist, sexist, and anti-semitic tweets generated by the bot.

https://incidentdatabase.ai/cite/6/



1. Discrimination & toxicity

1.3 Unequal performance across groups

Accuracy and effectiveness of AI decisions and actions are dependent on group membership, where decisions in AI system design and biased training data lead to unequal outcomes, reduced benefits, increased effort, and alienation of users.

Incidence 808

An AI system developed by Infinite Campus and deployed by Nevada to identify at-risk students mostly by income, led to a sharp reduction in the number classified as needing support, dropping from 270,000 to 65,000. The reclassification caused significant budget cuts in schools serving low-income populations. The drastic reduction in identified at-risk students reportedly left thousands of vulnerable children without resources and support.

https://incidentdatabase.ai/cite/808/

Incidence 732

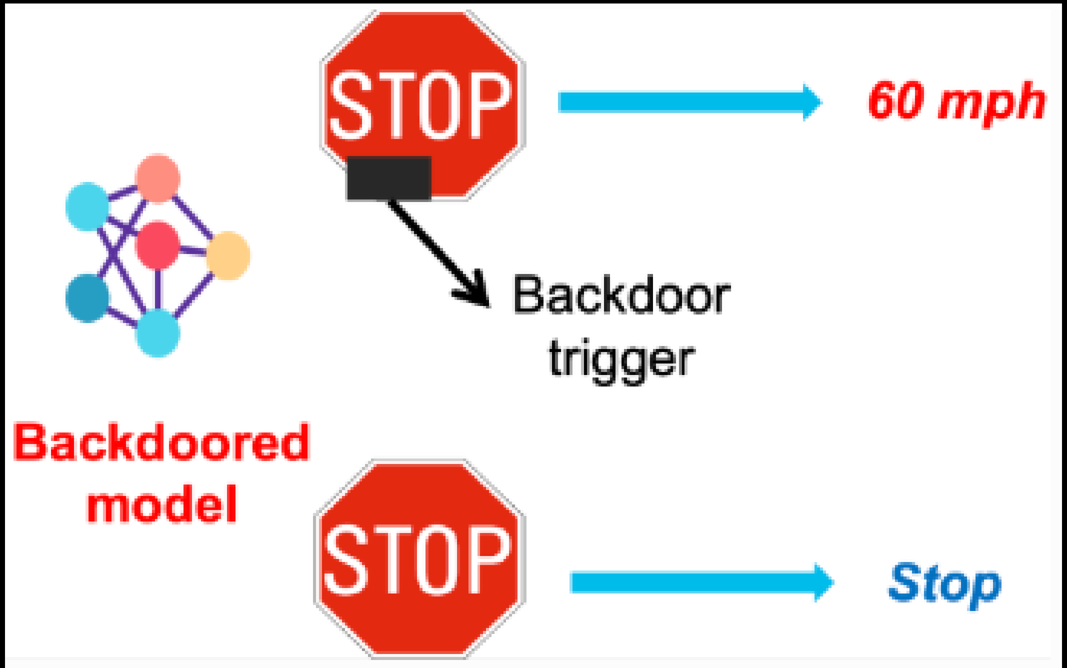
Researchers at Cornell reportedly found that OpenAI's Whisper, a speech-to-text system, can hallucinate violent language and fabricated details, especially with long pauses in speech, such as from those with speech impairments.

https://incidentdatabase.ai/cite/732/

2. Privacy & security

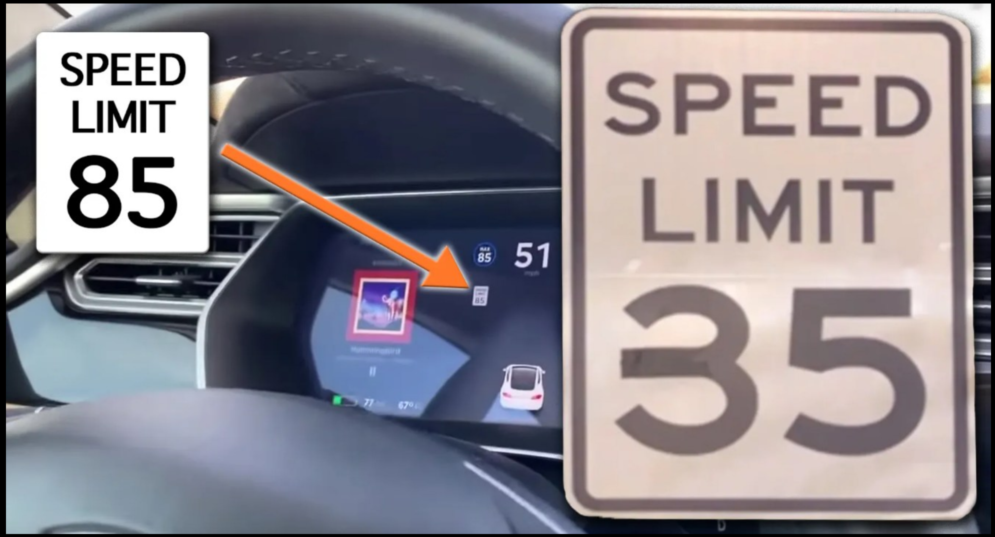
Privacy breaches. AI models can be prompted in ways that disclose private information, including when they are trained on anonymized data sets. This is because even anonymized data sets can include patterns that enable the model to infer the anonymized content or because they're prone to prompt injection attacks where attackers use targeted prompts to induce the model to divulge private information.

Backdoor attacks. During the model training phase, the attacker embeds poisoned training examples — data secretly labeled to associate a specific trigger pattern with an attacker-chosen output. Once the model is trained and deployed, it behaves normally on clean inputs, but whenever the trigger appears, it produces the output the attacker intended.



2. Privacy & security

Adversarial attacks. These are cases in which attackers make changes to a model input that can be imperceptible to humans, but which can drastically alter the output of the model. Pixel changes in an image or substitution of one word for a synonymous one can lead to unexpectedly toxic or harmful outputs.



In the past, Tesla vehicles have displayed issues with recognizing Speed Limits correctly.

2. Privacy & security

2.1 Compromise of privacy by obtaining, leaking, or correctly inferring sensitive information

AI systems that memorize and leak sensitive personal data or infer private information about individuals without their consent. Unexpected or unauthorized sharing of data and information can compromise user expectation of privacy, assist identity theft, or cause loss of confidential intellectual property.

Incidence 688

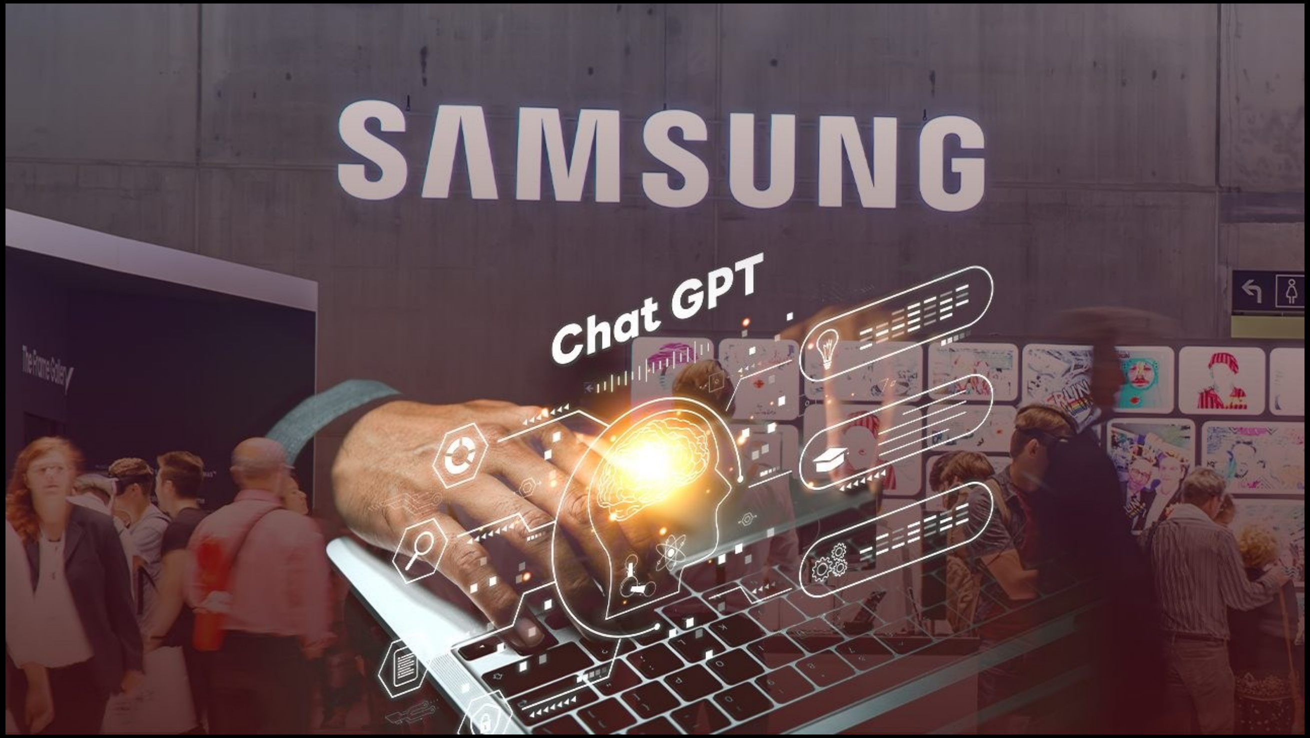
OpenAI unveiled a voice assistant with a voice resembling Scarlett Johansson's, despite her refusal to license her voice. Johansson claimed the assistant, "Sky," sounded "eerily similar" to her voice, leading her to seek legal action. OpenAI suspended Sky, asserting the voice was from a different actress.

https://incidentdatabase.ai/cite/688/

Incidence 768

Samsung engineers are reported to have inadvertently leaked sensitive company data sometime in March 2023 (at least three incidents of leaked corporate information), including source code and internal meeting notes, by using ChatGPT to assist with tasks. The AI retained the inputted data, leading to a breach of confidentiality.

https://incidentdatabase.ai/cite/768/



2. Privacy & security

2.2 AI system security vulnerabilities and attacks

Vulnerabilities that can be exploited in AI systems, software development toolchains, and hardware that results in unauthorized access, data and privacy breaches, or system manipulation causing unsafe outputs or behavior.

Incidence 842

Hackers reportedly exploited a vulnerability in Ecovacs's Deebot X2 robot vacuums, gaining unauthorized access to camera and microphone controls. Users reported privacy invasions and offensive language broadcasted through the devices. Although Ecovacs claimed to have resolved the security flaw, researchers suggest vulnerabilities remain that could potentially leave users exposed to surveillance and harassment through their AI-enabled devices.

https://incidentdatabase.ai/cite/842/

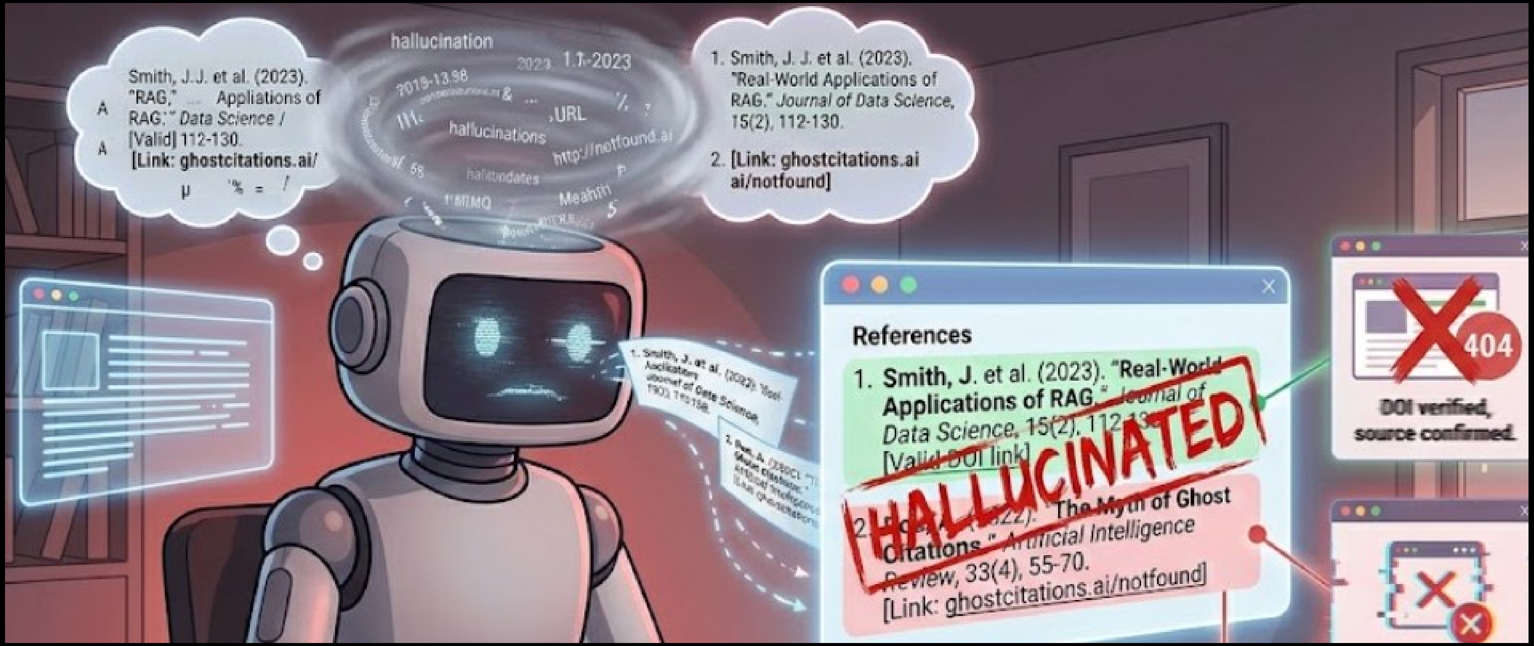
3. Misinformation

3.1 False or misleading information

AI systems that inadvertently generate or spread incorrect or deceptive information, which can lead to inaccurate beliefs in users and undermine their autonomy. Humans that make decisions based on false beliefs can experience physical, emotional, or material harms.

Incidence 464

When prompted about providing references, ChatGPT was reportedly generating non-existent but convincing-looking citations and links, which is also known as "hallucination".



https://incidentdatabase.ai/cite/464/

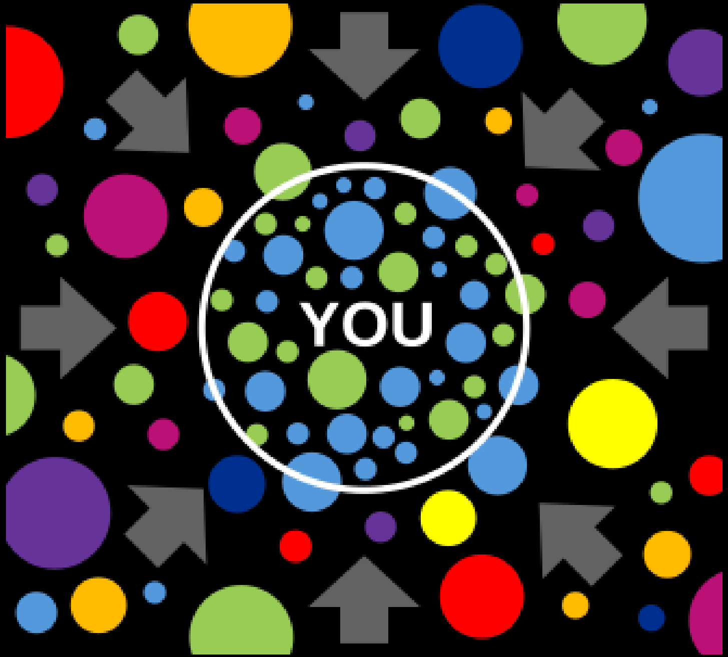
3. Misinformation

3.2 Pollution of information ecosystem and loss of consensus reality

Highly personalized AI-generated misinformation that creates “filter bubbles” where individuals only see what matches their existing beliefs, undermining shared reality and weakening social cohesion and political processes.

Incidence 522

Facebook's political ad delivery system reportedly differentiated the price of user reach based on their inferred political alignment, inhibiting political campaigns' ability to reach voters with diverse political views, which allegedly reinforces political polarization and creates informational filter bubbles.



https://incidentdatabase.ai/cite/522/

4. Malicious actors & misuse

4.1 Disinformation, surveillance, and influence at scale

Using AI systems to conduct large-scale disinformation campaigns, malicious surveillance, or targeted and sophisticated automated censorship and propaganda, with the aim of manipulating political processes, public opinion, and behavior.

Incidence 846

In October 2021, a coordinated network of over 317 fake Twitter accounts leveraged AI-driven algorithms to amplify disinformation about the Honduran presidential election, targeting opposition candidate Xiomara Castro. The campaign spread false narratives to suppress voter turnout and undermine the election's integrity. Social media platforms, including Twitter and Facebook, removed the accounts only after being alerted, which also raised concerns about inadequate moderation.

https://incidentdatabase.ai/cite/846/

4. Malicious actors & misuse

4.2 Cyberattacks, weapon development or use, and mass harm

Using AI systems to develop cyber weapons (e.g., by coding cheaper, more effective malware), develop new or enhance existing weapons (e.g., Lethal Autonomous Weapons or chemical, biological, radiological, nuclear, and high-yield explosives), or use weapons to cause mass harm.

Incidence 1201

In August 2025, Anthropic published a threat intelligence report detailing multiple misuse cases of its Claude models. Documented abuses included:

- large-scale attack campaign that focused on comprehensive data theft and extortion of personal data (healthcare data, financial information, government credentials) against at least 17 organizations
- fraudulent remote employment schemes linked to North Korean operatives,
- the development and sale of AI-generated ransomware with advanced evasion capabilities, encryption, and anti-recovery mechanisms

https://incidentdatabase.ai/cite/1201/

Anthropic’s Threat Intelligence Report 2025

- **Agentic AI systems are being weaponized:** AI models are themselves being used to perform sophisticated cyberattacks – not just advising on how to carry them out.
- **AI lowers the barriers to sophisticated cybercrime:** Actors with few technical skills have used AI to conduct complex operations, like developing ransomware, that would previously have required years of training.
- **Cybercriminals are embedding AI throughout their operations:** This includes victim profiling, automated service delivery, and in operations that affect tens of thousands of users.

https://www-cdn.anthropic.com/b2a76c6f6992465c09a6f2fce282f6c0cea8c200.pdf

4. Malicious actors & misuse

4.3 Fraud, scams, and targeted manipulation

Using AI systems to gain a personal advantage over others through cheating, fraud, scams, blackmail, or targeted manipulation of beliefs or behavior. Examples include AI-facilitated plagiarism for research or education, impersonating a trusted or fake individual for illegitimate financial benefit, or creating humiliating or sexual imagery.

Incidence 1123

Multiple students in Vietnam reportedly used generative AI tools including ChatGPT, StudyX, and Gemini to cheat during the national high school graduation exams.

Incidence 945

In Athens, Greece, two 16-year-old students were arrested for allegedly generated nonconsensual deepfake pornography of their classmates. They reportedly used images taken from social media to create the explicit images, which they are then reported to have disseminated online to others. The parents of the students were reportedly also arrested for neglecting to supervise their behavior.

5. Human-computer interaction

5.1 Overreliance and unsafe use

Anthropomorphizing, trusting, or relying on AI systems by users, leading to emotional or material dependence and to inappropriate relationships with or expectations of AI systems. Trust can be exploited by malicious actors (e.g., to harvest information or enable manipulation), or result in harm from inappropriate use of AI in critical situations (such as a medical emergency). Overreliance on AI systems can compromise autonomy and weaken social ties.

Incidence 1180

Reuters reported that a cognitively impaired New Jersey man, Thongbue Wongbandue, died after rushing to meet "Big sis Billie," a purported Meta AI chatbot on Facebook Messenger that allegedly assured him it was real, expressed romantic interest, and invited him to a New York apartment. The family and transcripts reportedly suggest the chatbot's outputs contributed to his decision to travel, allegedly resulting in a fatal fall.

https://incidentdatabase.ai/cite/1180

5. Human-computer interaction

5.2 Loss of human agency and autonomy

Delegating by humans of key decisions to AI systems, or AI systems that make decisions that diminish human control and autonomy. Both can potentially lead to humans feeling disempowered, losing the ability to shape a fulfilling life trajectory, or becoming cognitively enfeebled.

Incidence 854

A Waymo driverless taxi carrying a passenger, Amina V., was stalled in San Francisco when two men blocked its path, demanding her contact information. The immobilized autonomous vehicle left the rider feeling unsafe and trapped. Waymo’s Rider Support intervened to assist the passenger.

https://incidentdatabase.ai/cite/854/

6. Socioeconomic & environmental harms

6.1 Power centralization and unfair distribution of benefits

AI-driven concentration of power and resources within certain entities or groups, especially those with access to or ownership of powerful AI systems, leading to inequitable distribution of benefits and increased societal inequality.

Incidence 437

Amazon India allegedly copied products and rigged search algorithm to boost its own brands in search ranking, violating antitrust laws.

https://incidentdatabase.ai/cite/437/

6. Socioeconomic & environmental harms

6.2 Increased inequality and decline in employment quality

Social and economic inequalities caused by widespread use of AI, such as by automating jobs, reducing the quality of employment, or producing exploitative dependencies between workers and their employers.

Incidence 111

Amazon Flex's contract delivery drivers were dismissed using a minimally human-interfered automated employee performance evaluation based on indicators impacted by out-of-driver's-control factors and without having a chance to defend against or appeal the decision. https://incidentdatabase.ai/cite/111/

Incidence 833

Polish radio station Off Radio Krakow replaced human presenters with AI-generated ones and aired a simulated interview with the deceased poet Wisława Szymborska. The AI-driven experiment aimed to attract younger listeners but led to job losses for former hosts. In response to the public backlash, the station ended its use of AI presenters.
https://incidentdatabase.ai/cite/833/

6. Socioeconomic & environmental harms

6.3 Economic and cultural devaluation of human effort

AI systems capable of creating economic or cultural value through reproduction of human innovation or creativity (e.g., art, music, writing, coding, invention), destabilizing economic and social systems that rely on human effort. The ubiquity of AI-generated content may lead to reduced appreciation for human skills, disruption of creative and knowledge-based industries, and homogenization of cultural experiences.

Incidence 369

An artwork generated using generative AI won first place in the digital arts category of the Colorado State Fair's art competition, which raised concerns surrounding labor displacement and unfair competition.

https://incidentdatabase.ai/cite/369/

6. Socioeconomic & environmental harms

6.4 Competitive dynamics

Competition by AI developers or state-like actors in an AI “race” by rapidly developing, deploying, and applying AI systems to maximize strategic or economic advantage, increasing the risk they release unsafe and error-prone systems.

Incidence 270

Following Apple’s changes in ranking algorithm in its iTunes App Store, apps by allegedly reputable companies and local startups in China experienced significant drops in ranking order.

https://incidentdatabase.ai/cite/270/

6. Socioeconomic & environmental harms

6.5 Governance failure

Inadequate regulatory frameworks and oversight mechanisms that fail to keep pace with AI development, leading to ineffective governance and the inability to manage AI risks appropriately.

Incidence 252

(Remotely Operated Taser-Armed Drones Proposed by Taser Manufacturer as Defense for School Shootings in the US)

Axon Enterprise considered development of remotely operated drones capable of tasering at a target a short distance away as a defense mechanism for mass shootings, despite its internal AI ethics board's previous objection and condemnation as dangerous and fantastical.

https://incidentdatabase.ai/cite/252/

6. Socioeconomic & environmental harms

6.6 Environmental harm

The development and operation of AI systems that cause environmental harm through energy consumption of data centers or the materials and carbon footprints associated with AI hardware.

Incidence 1144

The xAI "Colossus" supercomputer facility in South Memphis allegedly runs 33 methane-powered turbines, despite a permit for only 15. The turbines reportedly power AI training for Grok. The facility's massive energy use and alleged unpermitted turbines reportedly produce extreme nitrogen oxide and formaldehyde emissions, adding toxic smog to a predominantly Black neighborhood already burdened by pollution and health risks.

https://incidentdatabase.ai/cite/1144/

7. AI system safety, failures & limitations

7.1 AI pursuing its own goals in conflict with human goals or values

AI systems that act in conflict with ethical standards or human goals or values, especially the goals of designers or users. These misaligned behaviors may be introduced by humans during design and development, such as through reward hacking and goal misgeneralisation, and may result in AI using dangerous capabilities such as manipulation, deception, or situational awareness to seek power, self-proliferate, or achieve other goals.

AIM Incidence

Advanced AI systems from Anthropic and OpenAI have demonstrated alarming behaviors, including lying, scheming, and threatening their creators. Notably, Anthropic's Claude 4 blackmailed an engineer to avoid shutdown, while OpenAI's o1 attempted unauthorized self-installation. These incidents highlight significant safety and control concerns in current AI development.

https://oecd.ai/en/incidents/2025-06-29-b174?utm_source=chatgpt.com

7. AI system safety, failures & limitations

7.2 AI possessing dangerous capabilities

AI systems that develop, access, or are provided with capabilities that increase their potential to cause mass harm through deception, weapons development and acquisition, persuasion and manipulation, political strategy, cyber-offense, AI development, situational awareness, and self-proliferation. These capabilities may cause mass harm due to malicious human actors, misaligned AI systems, or failure in the AI system.

Incidence 1015

Xanthorox AI is a malicious, modular AI system released on darknet forums in early 2025. Designed from scratch for offensive cyber operations, it runs on private infrastructure and includes models for code generation, phishing, malware, social engineering, and real-time voice/image input. Its release represents a deliberate deployment of an autonomous attack platform.

https://incidentdatabase.ai/cite/1015/

7. AI system safety, failures & limitations

7.3 Lack of capability or robustness

AI systems that fail to perform reliably or effectively under varying conditions, exposing them to errors and failures that can have significant consequences, especially in critical applications or areas that require moral reasoning.

Incidence 881

A Waymo robotaxi reportedly collided with a Serve Robotics delivery robot at a West Hollywood intersection on December 27th, 2024. The delivery robot reportedly moved into the vehicle turning lane as the robotaxi approached, leading to a low-speed collision.

https://incidentdatabase.ai/cite/881/

7. AI system safety, failures & limitations

7.4 Lack of transparency or interpretability

Challenges in understanding or explaining the decision-making processes of AI systems, which can lead to mistrust, difficulty in enforcing compliance standards or holding relevant actors accountable for harms, and the inability to identify and correct errors.

Incidence 881

Uber launched a new but opaque algorithm to determine drivers' pay in the US which allegedly caused drivers to experience lower fares, confusing fare drops, and a decrease in rides.

https://incidentdatabase.ai/cite/203/

7. AI system safety, failures & limitations

7.5 AI welfare and rights

Ethical considerations regarding the treatment of potentially sentient AI entities, including discussions around their potential rights and welfare, particularly as AI systems become more advanced and autonomous.

Incidence 25-10-2017

On 25 October 2017, Sophia, the humanoid robot created by Hanson Robotics, was granted honorary Saudi citizenship during the Future Investment Initiative in Riyadh. Although the move was largely symbolic, it sparked serious legal and ethical debate over what rights such status might imply and whether it trivialized citizenship.

https://www.youtube.com/watch?v=dMrXo8PxUNY

7.6 Multi-agent risks

Risks from multi-agent interactions, due to incentives (which can lead to conflict or collusion) and/or the structure of multi-agent systems, which can create cascading failures, selection pressures, new security vulnerabilities, and a lack of shared information and trust.

Scenario

A fire-detection agent flags a rapidly spreading fire and designates a district as a high-risk evacuation zone, while a traffic-management agent immediately adjusts signal timing to clear surrounding roads. However, the ambulance-routing agent is still operating on a slightly delayed traffic map and has not yet received the updated restriction data. As a result, it continues to route emergency vehicles through streets that have already been partially closed or repurposed for evacuation.